Monash FIT3143 Parallel Computing | Applied #2: GPU/NPU and Exponential Growth

GPU Image Rotation and the Ethics of Scaling HPC for AI

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Task 1: GPU design

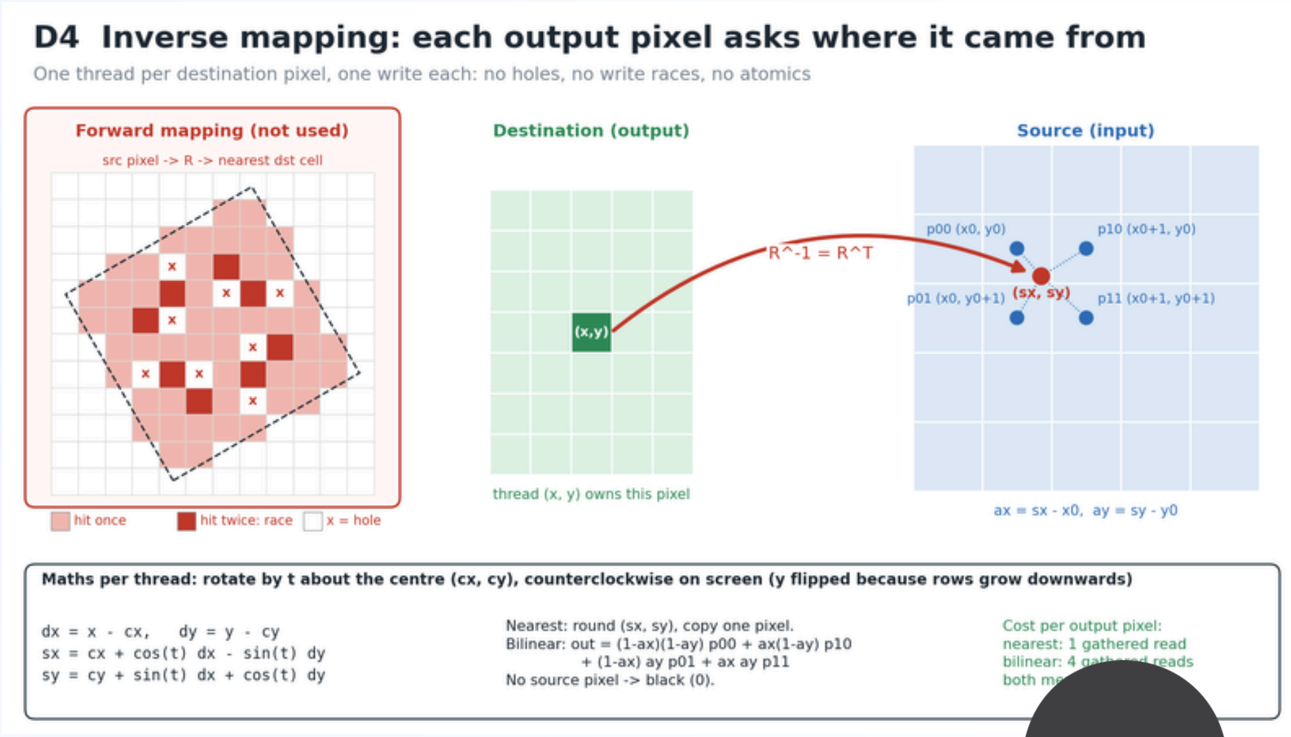
Task 2: Ethics of HPC

Limits and Q&A

Presenter: Taabish

Task 1: 33 Million Independent Pixels

- 8K = 7680 x 4320 = 33,177,600 output pixels
- Each pixel is independent: a data-parallel workload
- Inverse mapping: one thread per output pixel
- One writer per pixel, no holes

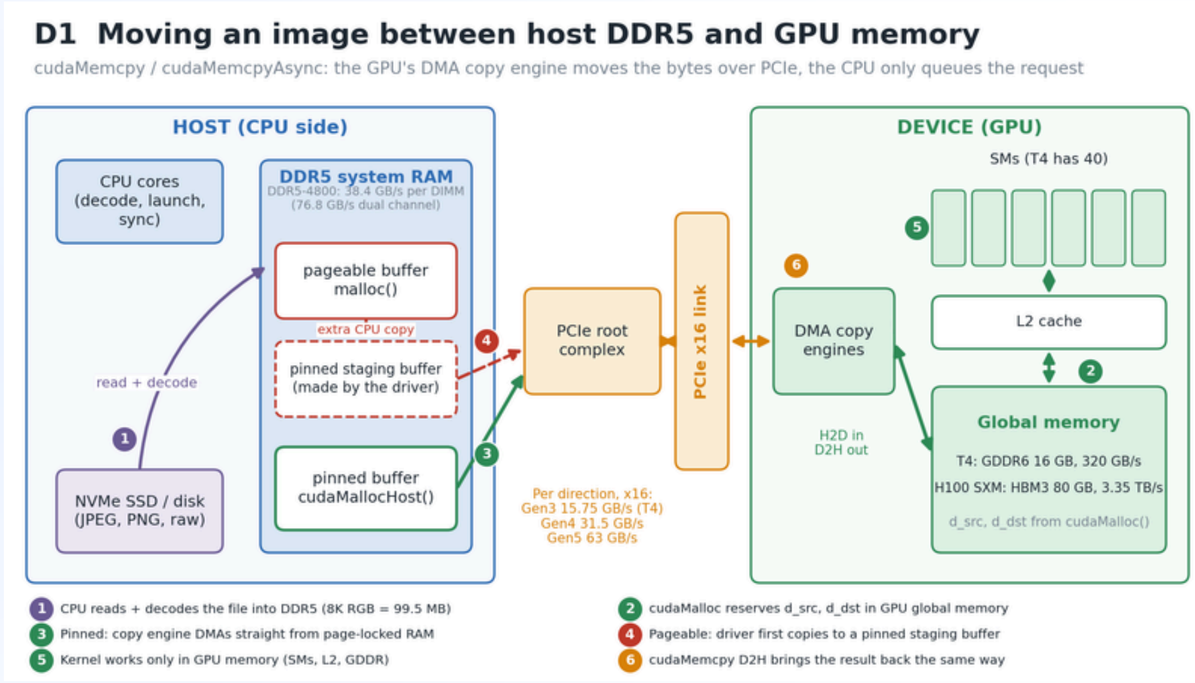




Presenter: Taabish

1a. Host to Device and Back

- Pageable: extra staging copy (4.6 GB/s)
- Pinned: DMA straight over PCIe
- Pinned H2D measured: 12.3 GB/s; T4 GDDR6: 320 GB/s (NVIDIA, n.d.-d)
- Unified Memory and zero-copy move data on demand (NVIDIA, 2026a, 2026b)





Sources: NVIDIA (n.d.-d, 2026a, 2026b). Measured on Tesla T4.

Presenter: Taabish

1b. CUDA Model: host, device, <<<grid, block>>>

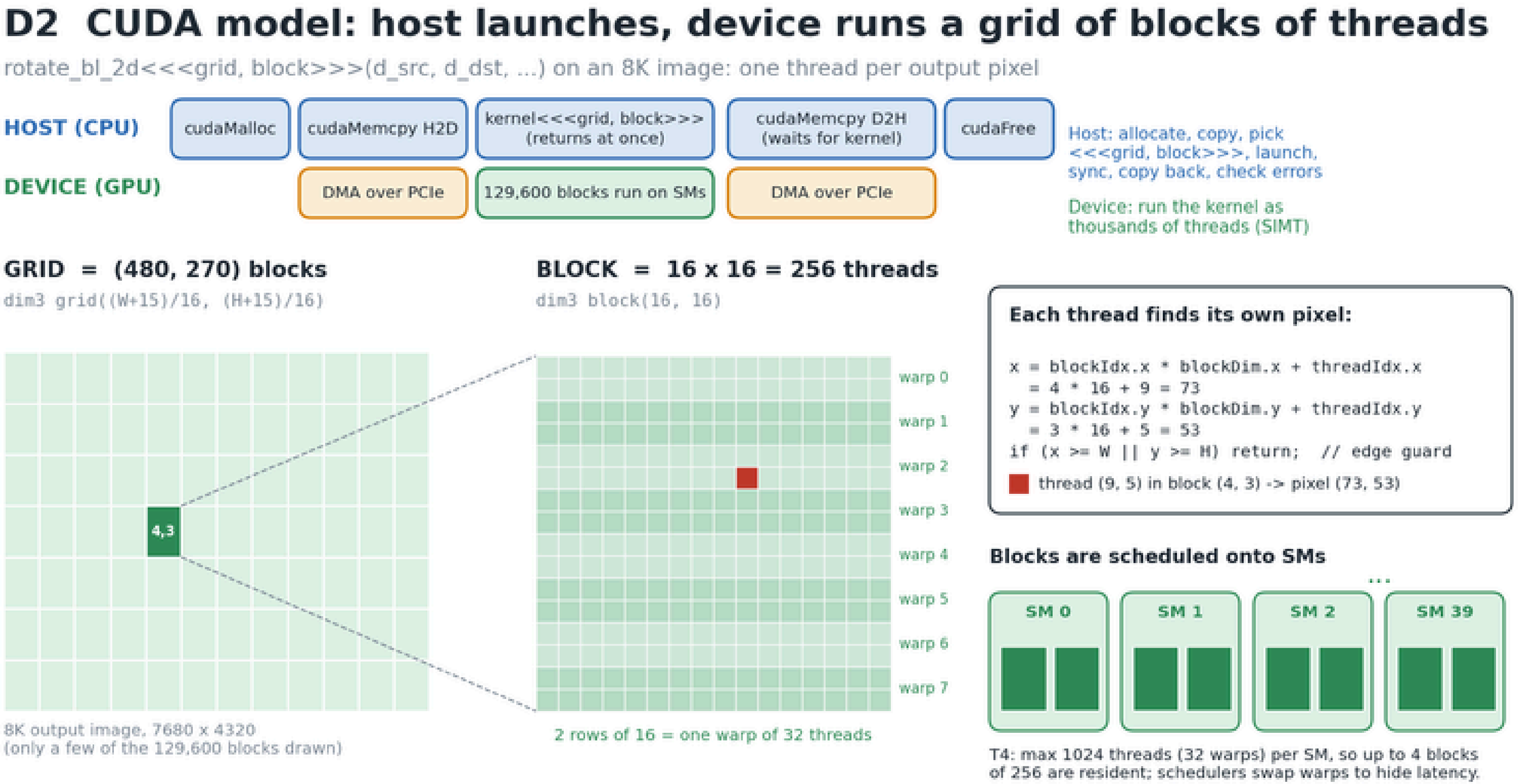
- Host (CPU): allocate, copy, launch async, sync, check errors
- Device (GPU): SIMT threads in warps of 32
- Block 16 x 16 = 256 threads = 8 warps
- Grid 480 x 270 = 129,600 blocks for 8K



Sources: NVIDIA (2026b).

Presenter: Taabish

1b. CUDA model: grid, block, thread



Sources: NVIDIA (2026b).

1b. Kernel and launch (from rotate_cuda.cu)

Presenter: Taabish

YOUR SCREENSHOT 1

Kernel: rotate_nn_2d

rotate_cuda.cu, lines 145 to 158

YOUR SCREENSHOT 2

Grid size and launch

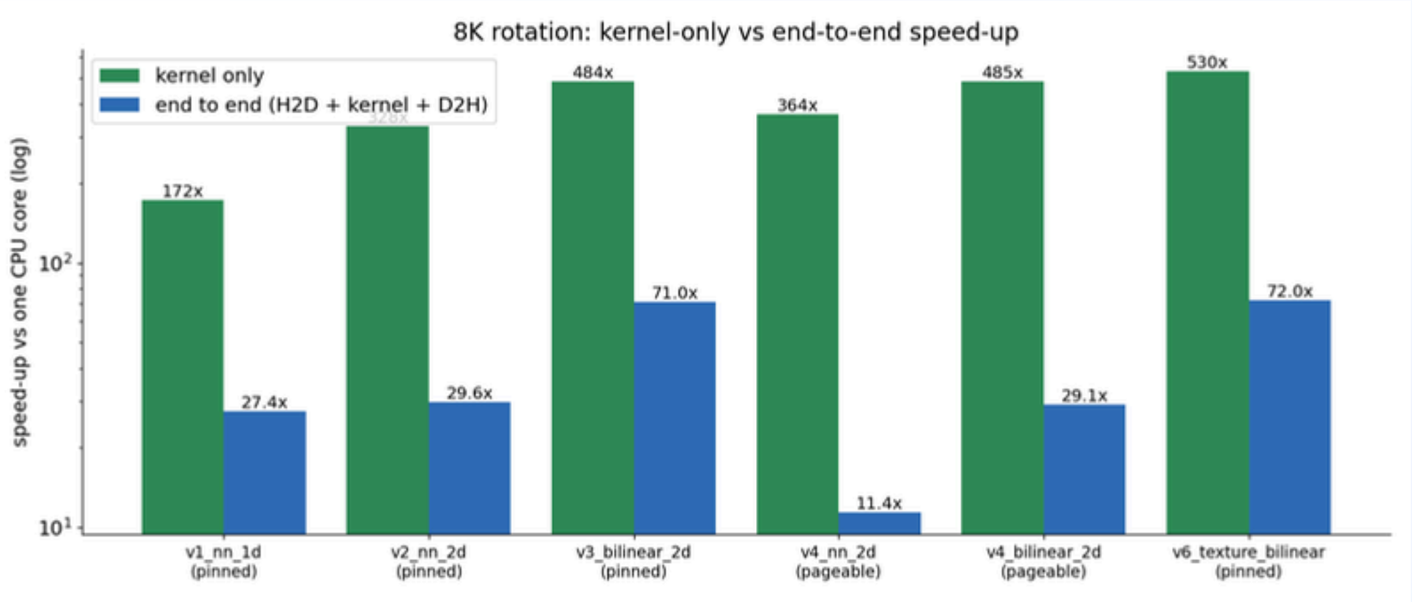
rotate_cuda.cu, lines 360 to 383

- Each thread owns one output pixel and reads its source with R transpose, so there are no holes and no write races
- Guard drops extra threads; grid is rounded up
- Launch is async, so we check the error right after
- Simplified from the file in our submission

Sources: rotate_cuda.cu in our submission zip.

Presenter: Taabish

1b. Each Feature, Measured



- v1 1D grid, v2 2D blocks, v3 bilinear, v4 pageable, v5 streams, v6 texture
- 2D grid beat 1D: 1.55 vs 2.96 ms
- Amdahl cap: 32.6x (next slide)

328x
kernel only

29.6x
end to end

91%
of the time is copies

Tesla T4, 8K at 30 degrees, v2 nearest, mean of 10 runs. CPU baseline: one core.

Presenter: Taabish

1b. Six versions, and what limits us

Six versions

- v1 1D grid
- v2 2D 16 x 16 blocks
- v3 bilinear
- v4 pageable memory
- v5 streams
- v6 texture object

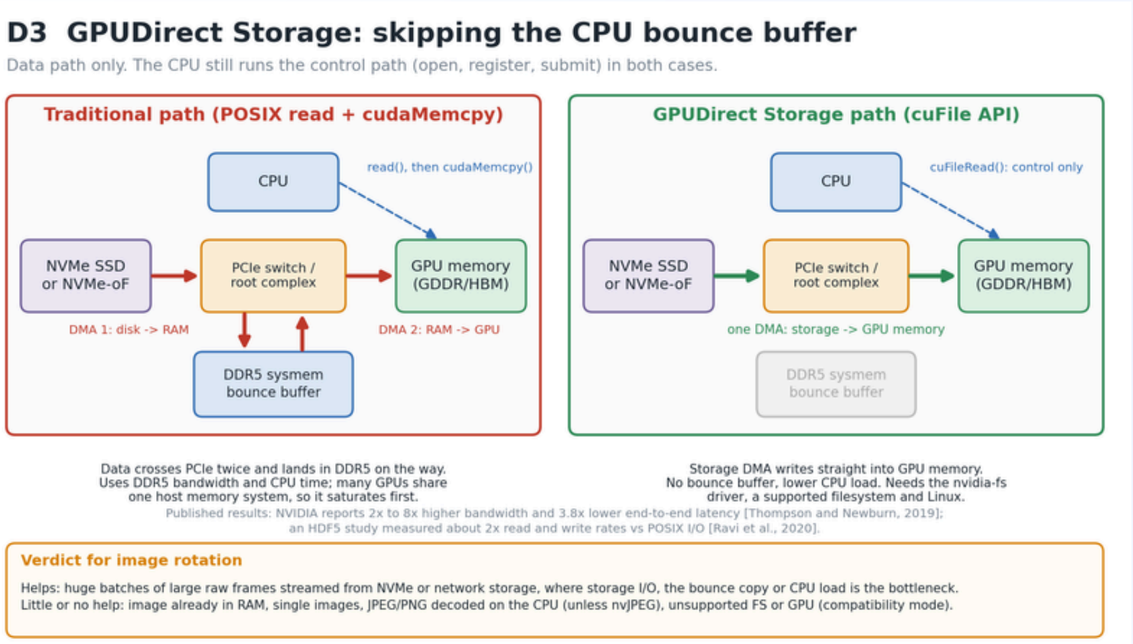
Why the speed-up stops

- Amdahl bound with a zero-time kernel: 509.3 / (8.06 + 7.58) = 32.6x, so 29.6x is 91% of it
- All CPU cores (OpenMP): Colab's 2 vCPUs take 470 ms, so the T4 is still 27x end to end
- A 14-thread laptop takes 14.5 ms, faster than the T4's 17.2 ms

v2 (nearest, 16 x 16), 8K at 30 degrees, Tesla T4, mean of 10 runs.

1c. GPUDirect Storage: helps the bulk pipeline, not one photo

Presenter: Taabish



Helps

- Bulk raw frames, or JPEGs decoded on the GPU (nvJPEG)
- When I/O is the bottleneck (Thompson & Newburn, 2019)

Does not help

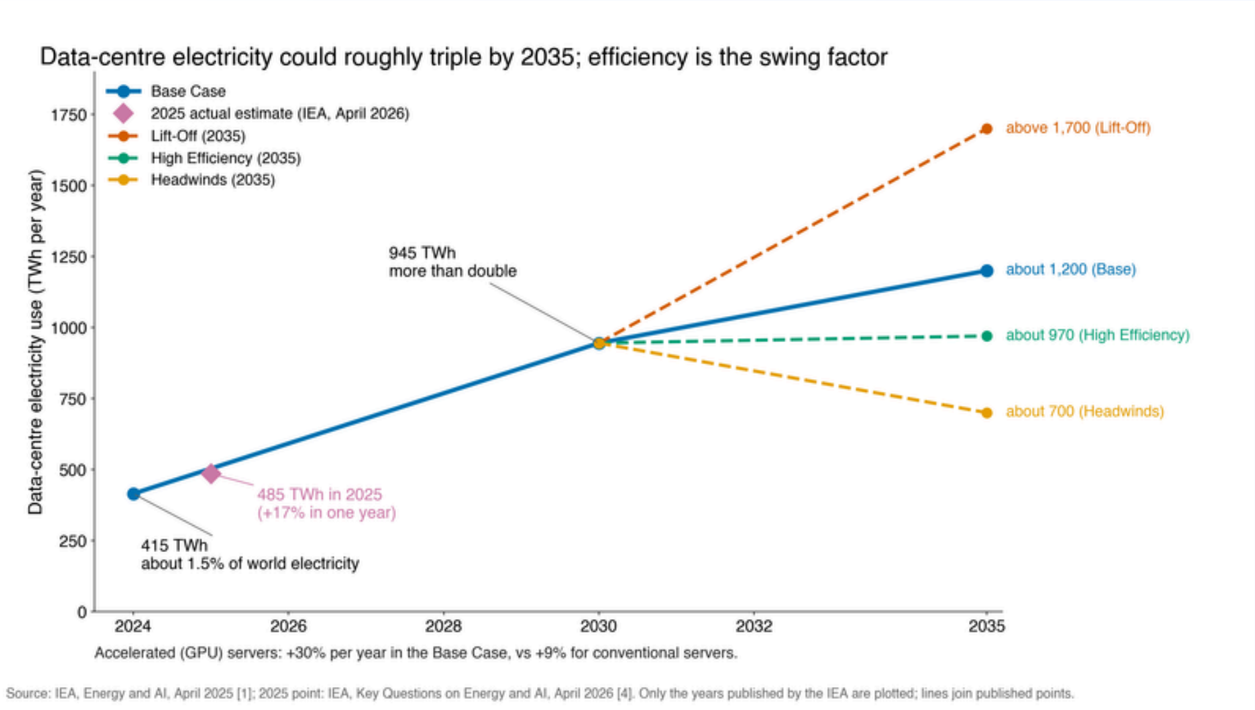
- One photo
- JPEGs decoded on the CPU
- Images already in RAM
- Needs nvidia-fs and O_DIRECT: we could not test it on Colab (NVIDIA, n.d.-a)

Not tested: Colab has no nvidia-fs module

Sources: Thompson & Newburn (2019); NVIDIA (n.d.-a).

Presenter: Erwyna

2a. Energy: data centre demand keeps climbing

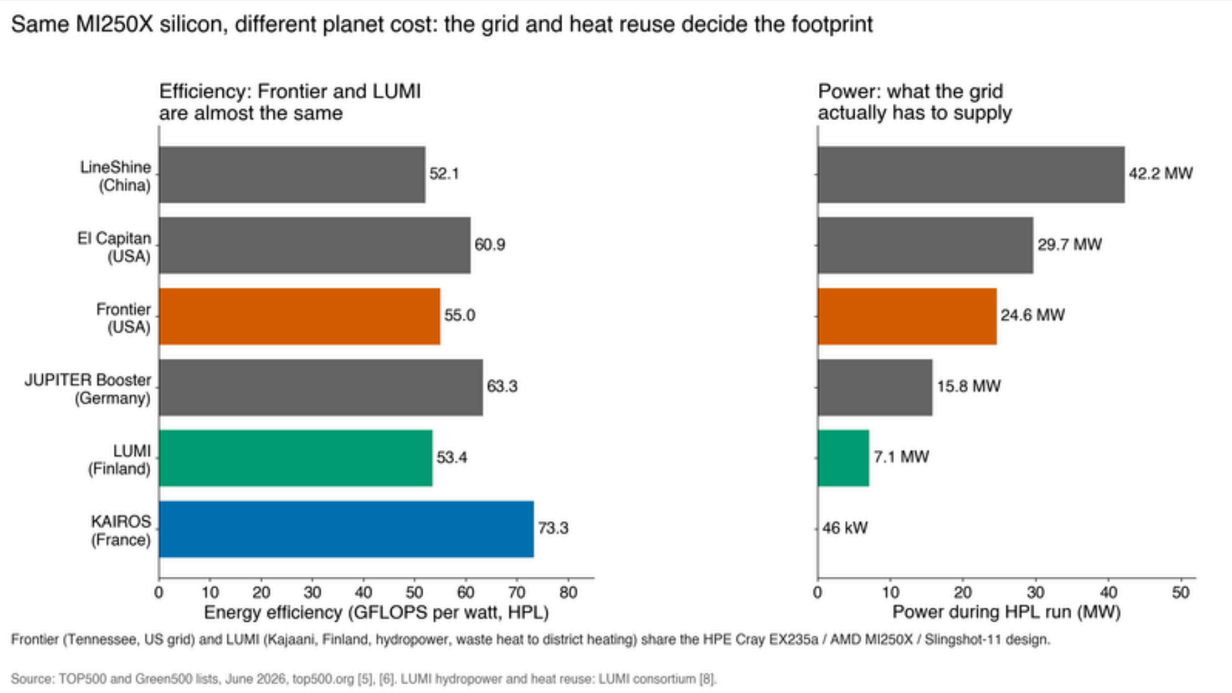


- 415 TWh in 2024
- 945 TWh by 2030, about 1,200 TWh by 2035
- GPU servers drive the growth (IEA, 2025)

Sources: International Energy Agency (IEA, 2025).

2a. Same chip, different grid

Presenter: Erwyna



- Frontier: 55.0 GFLOPS/W
- LUMI: 53.4 GFLOPS/W
- LUMI: 100% hydropower
- Waste heat warms a town

Carbon per FLOP depends on the grid, not only the chip

Sources: TOP500 (2026); LUMI consortium (2023).

Presenter: Erwyna

2a. Malaysia lens and a myth check

Johor, Malaysia

- 79% fossil grid (Ember, 2026)
- 10 MW to 1.3 GW in 3 years (Loo, 2025)
- Up to 31% of Peninsular electricity by 2035 (Bernama, 2026)
- No new water-hungry Tier 1 and 2 sites (Shadiqe, 2025)

Kajaani, Finland (LUMI)

- 100% hydropower
- Cold climate cooling
- Waste heat warms the town (LUMI consortium, 2023)

Myth: "AI out-emits aviation"

It is a 2030 projection. Today: about 180 Mt CO2

Sources: Ember (2026); Loo (2025); Bernama (2026); LUMI consortium (2023); Jamison et al. (2025); Giles (2025); IEA (2025).

Presenter: Erwyna

2b. Data Governance, Security and Fairness

Governance

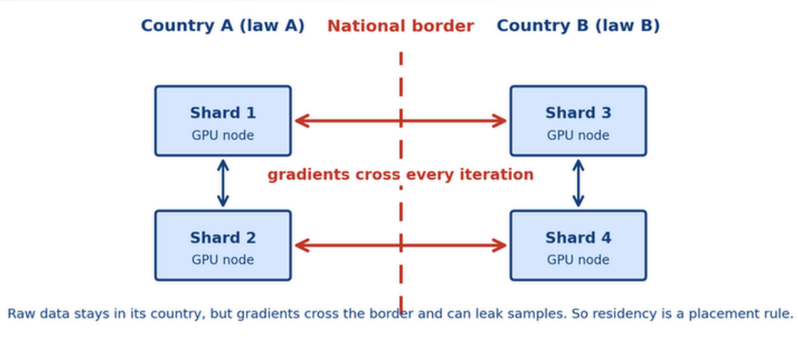
- 75% of AI supercomputer power is in the US (Pilz et al., 2025)
- Gradients can leak data (Zhu et al., 2019), so residency is a scheduling rule

Security: dual-use

- Same chips, hospital or military model: dual-use
- US limits exports; Malaysia now requires permits (BIS, 2022; MITI, 2025)

Fairness

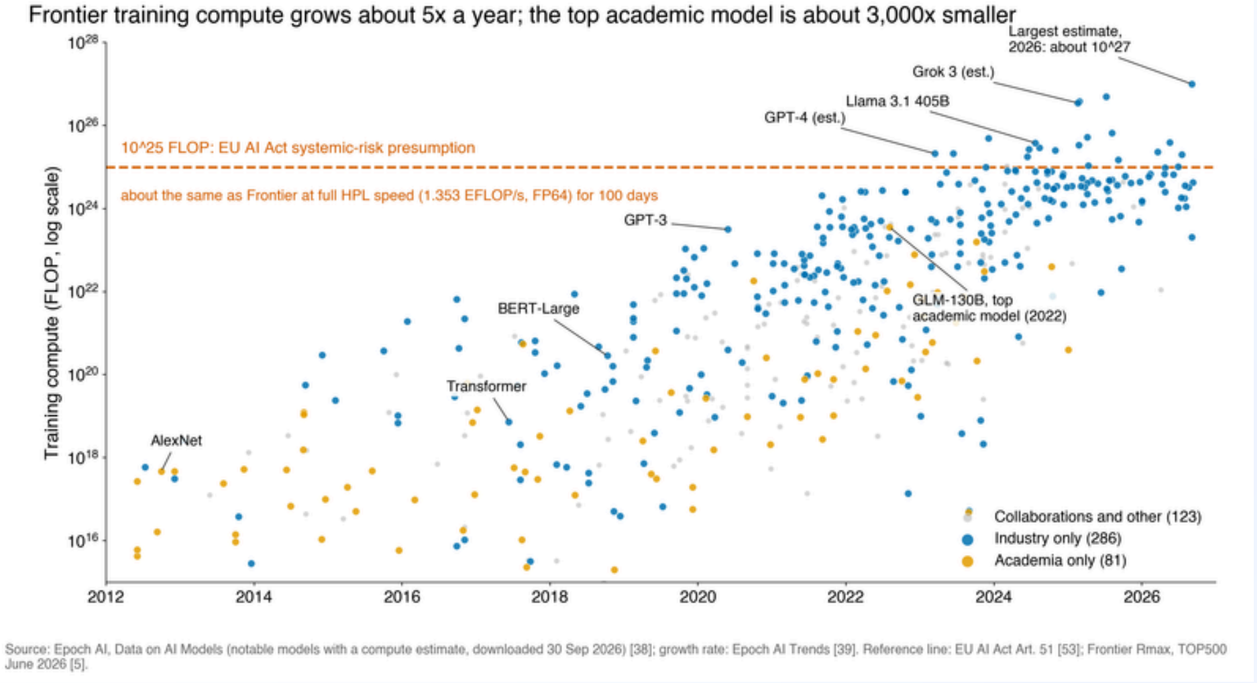
- Malay is 0.086% of web text
- Tamil takes about 10 tokens per word (Common Crawl, 2026; Ahuja et al., 2023)



Sources: Pilz et al. (2025); Zhu et al. (2019); BIS (2022); MITI (2025); Common Crawl (2026); Ahuja et al. (2023).

Presenter: Erwyna

2c. The Compute Divide



Source: Epoch AI, Data on AI Models (notable models with a compute estimate, downloaded 30 Sep 2026) [38]; growth rate: Epoch AI Trends [39]. Reference line: EU AI Act Art. 51 [53]; Frontier Rmax, TOP500 June 2026 [5].

Only companies can audit frontier models
NCI Gadi: 776 GPUs, demand about 3x allocation (NCI, n.d.-a, n.d.-b)

>90%

of notable models: industry

5x / year

frontier compute; top academic model about 3,000x lower

1 to 8 GPUs

for most academics

Sources: Stanford HAI (2026); Epoch AI (2026); Khandelwal et al. (2025); NCI (n.d.-a, n.d.-b).

2d. Our SCALE checklist

Presenter: Erwyna

Each step is a Slurm setting and a reported number

S Size right: stop adding GPUs when parallel efficiency drops

C Carbon-aware: up to 19% less CO2 for jobs that can wait

A Account every joule: energy per job, PUE reported

L Lawful, local data: keep data in its country

E Equitable access: fair-share queues

UNESCO, OECD, AIGE: values. EU: 10^25 FLOP, PUE above 500 kW

Sources: UNESCO (2021); OECD (2024); MOSTI (2024); European Parliament & Council (2024); Wiesner et al. (2021).

Presenter: Erwyna

Limits, Future Work and Takeaways

Task 1

- Limits: Colab has only 2 vCPUs, shared VM, no GDS test
- Next: nvJPEG + GDS + streams on a GDS server
- Takeaway: memory bound, so PCIe sets the ceiling

Task 2

- Limits: model energy figures are estimates (Epoch AI, 2026); carbon-aware gains only 1 to 2% at peak hours (Radovanovic et al., 2023)
- Ours, measured: 0.71 J per 8K image on the T4 (41 W for 17.2 ms)
- Takeaway: a FLOP's cost depends on where and for whom it runs

Sources: Epoch AI (2026); Radovanovic et al. (2023).

Presenter: Erwyna

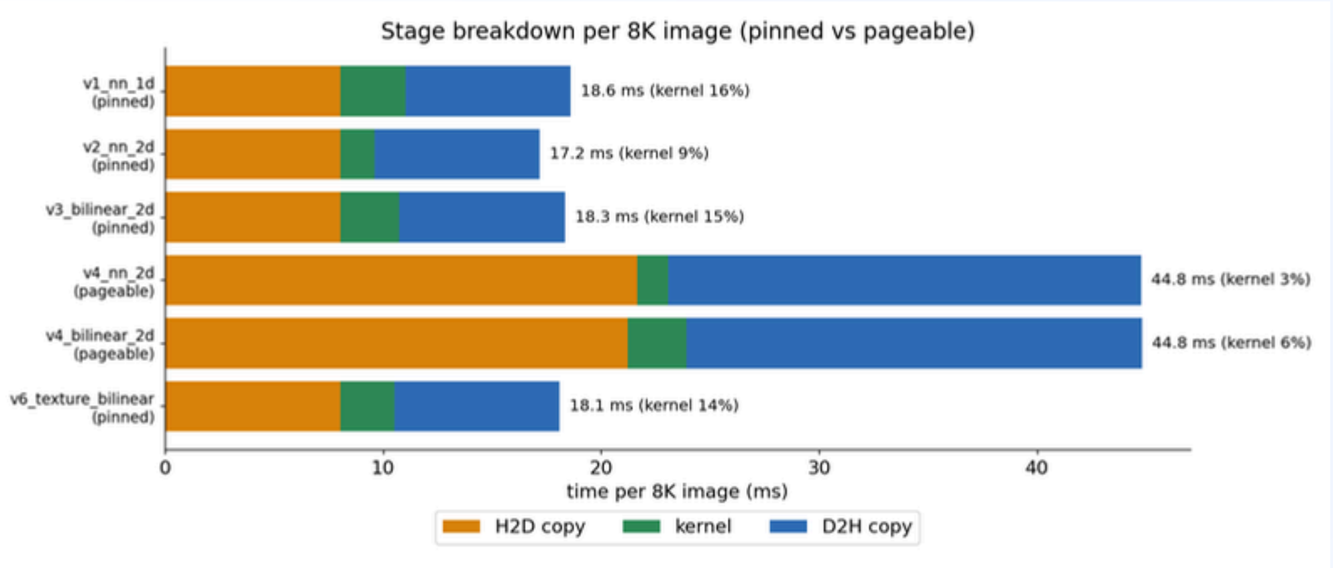
Q&A

Questions?

Erwyna Soo Wen Xin and Taabish Farooq Bhat

A1. Where the time goes (8K, 30 degrees)

APPENDIX: extra material for Q&A, not part of the 7-minute talk

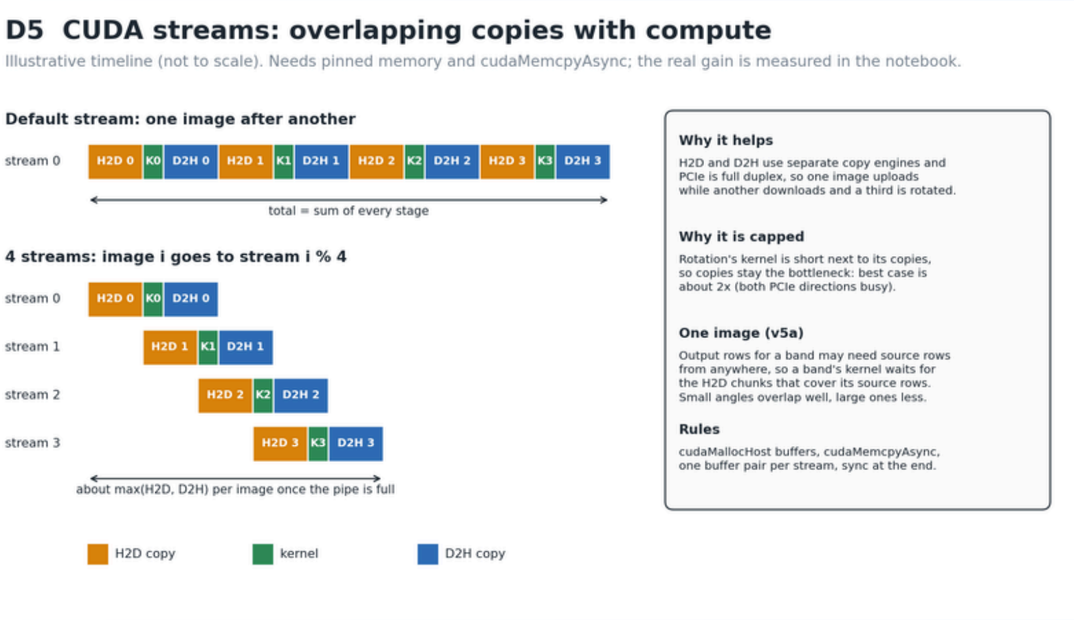


- H2D 8.06 ms
- Kernel 1.55 ms
- D2H 7.58 ms
- Total 17.19 ms
- CPU, one core: 509.3 ms
- Mismatch: 0 pixels

v2 nearest, pinned, Tesla T4, mean of 10 runs.

A2. Pinned vs pageable, and streams

APPENDIX: extra material for Q&A, not part of the 7-minute talk



YOUR SCREENSHOT 3

Streams loop with cudaMemcpyAsync

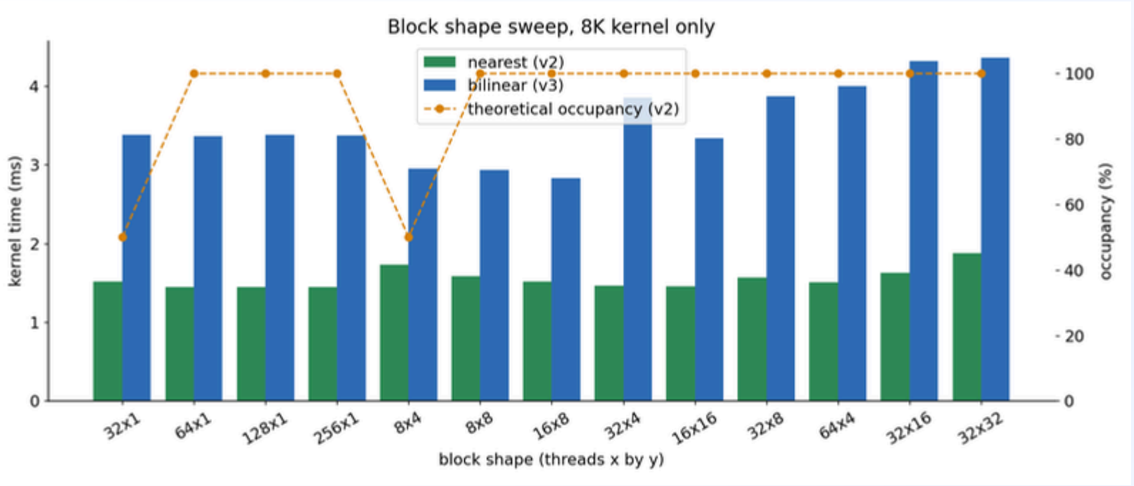
rotate_cuda.cu, lines 803 to 815

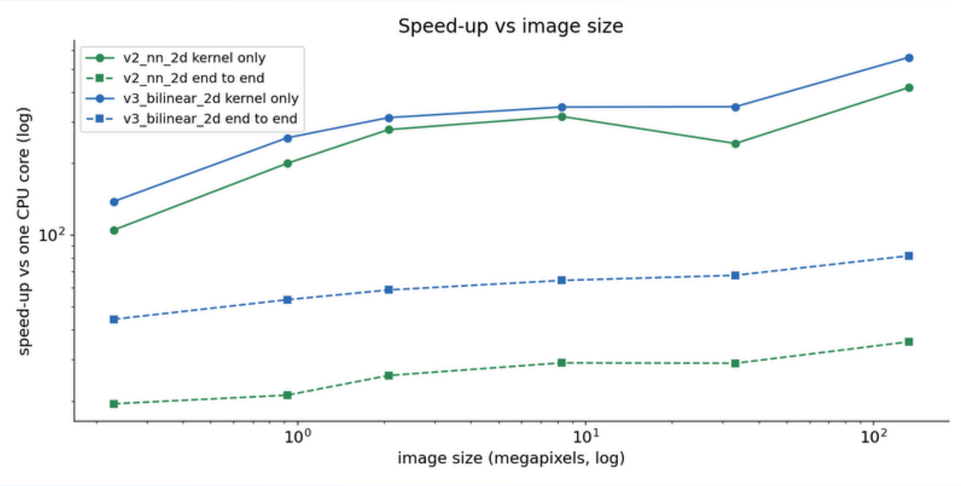
- Pageable 4.6 GB/s vs pinned 12.3 GB/s
- Streams: 1.8x on 8 images, 1.36x on one image

Image i goes to stream i % 4. Async copies need pinned memory. Source: rotate_cuda.cu in our submission zip.

A3. Block sweep and image size sweep

APPENDIX: extra material for Q&A, not part of the 7-minute talk





- Nearest: block shape barely matters
- Bilinear: 16 x 8 is fastest (2.83 ms)
- Bigger images: bigger speed-up

Run to run, the v2 kernel varied from 1.45 to 2.11 ms; copies stayed at 8.06 ms.

A4. How we timed it

APPENDIX: extra material for Q&A, not part of the 7-minute talk

YOUR SCREENSHOT 4

CUDA events around H2D, kernel and D2H

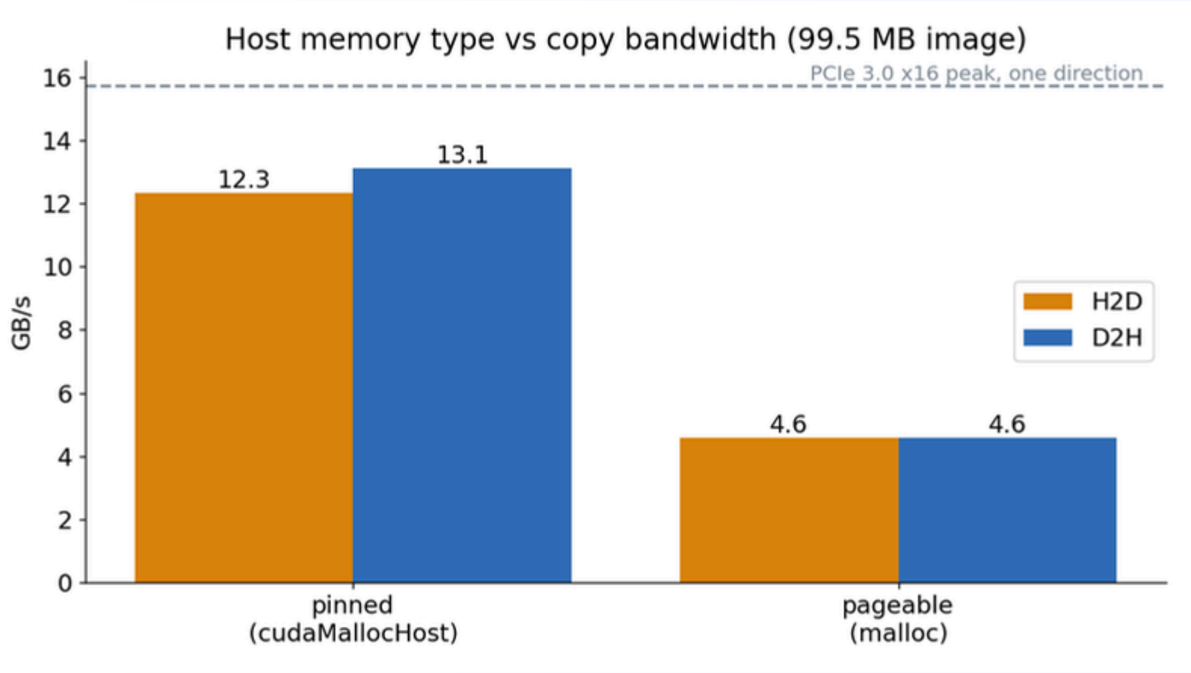
rotate_cuda.cu, lines 440 to 450

- Each stage timed on the GPU clock with events
- Warm-up run discarded, then mean of 10 runs
- --fmad=false: v1 to v5 match the CPU exactly

v6 texture differs in 257,964 bytes (0.26%), each by at most 1. Source: rotate_cuda.cu in our submission zip.

A5. Why PCIe is the narrow pipe

APPENDIX: extra material for Q&A, not part of the 7-minute talk



- PCIe 3.0 x16 (T4): 15.75 GB/s per direction
- Measured pinned copy: 12.3 GB/s
- T4 GDDR6: 320 GB/s
- About 25 FLOP per byte: rotation is memory bound
- PCIe is about 20x narrower, so copies are 91% of the time

Sources: PCI-SIG (2019); NVIDIA (n.d.-d). Roofline: T4 balance is about 25 FLOP per byte.

A6. GDS in practice

APPENDIX: extra material for Q&A, not part of the 7-minute talk

- Call order: cuFileDriverOpen, cuFileHandleRegister (file opened with O_DIRECT), cuFileBufRegister (GPU buffer), cuFileRead
- The CPU issues the calls; data DMAs NVMe to GPU with no host bounce buffer
- Needs nvidia-fs, O_DIRECT, aligned buffers, and a supported file system on local NVMe or NVMe-oF
- Without these, cuFile falls back to a host bounce buffer, which is what Colab would do

Sources: NVIDIA (n.d.-a, n.d.-b, n.d.-c).

A7. SIMD vs SIMT vs NPU

APPENDIX: extra material for Q&A, not part of the 7-minute talk

SIMD (CPU)	One instruction on a wide register, lanes in lockstep. Divergence is masked, which wastes lanes.
SIMT (GPU)	Warps of 32 threads run one instruction, each with its own index. Divergent branches serialise inside the warp.
NPU	Fixed matrix-multiply units (systolic array) for low-precision inference: high TOPS per watt, little flexibility. Rotation is a gather, so it maps poorly.

Sources: Lindholm et al. (2008); Jouppi et al. (2017).

A8. References, Task 1 (APA 7)

APPENDIX: extra material for Q&A, not part of the 7-minute talk

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A9. References, Task 2 (APA 7), part 1

APPENDIX: extra material for Q&A, not part of the 7-minute talk

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A9. References, Task 2 (APA 7), part 2

APPENDIX: extra material for Q&A, not part of the 7-minute talk

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